

Effect of brief mindfulness and relaxation inductions on anxiety, affect and brain activation in athletes

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ABSTRACT

The mindfulness-based intervention and psychological skills training are often used for maintaining the mental health or reducing undesirable mental states in athletes. However, their differences in acute effects on mental health and underlying neural mechanism are not well understood. Therefore, the purpose of the current study was to examine the differential effects of brief mindfulness induction (MI) and relaxation induction (RI) on state anxiety, affect and brain activation. Thirty-five track and field athletes were recruited for this study. Using a within-subjects crossover design, participants underwent three conditions that incorporated two 30-min experimental conditions (i.e., MI or RI) and a control condition. State anxiety and affect were assessed before and after intervention, and brain activation (i.e., theta, alpha bands) were recorded by electroencephalography (EEG) during each 30-min condition. Repeated measures analysis of variance revealed that MI and RI similarly reduced state anxiety and negative affect from pre-test to post-test compared to the control condition. In terms of positive affect, there were no significant differences among the three conditions across times. Furthermore, participants exhibited higher frontal theta power during the MI and RI than control condition, whereas no differences in alpha power were observed among conditions. The current study provides initial evidence from an electrophysiological perspective that brief MI and RI both improve the negative psychological states in individual sport athletes through similar neural mechanisms. Nevertheless, the moderating effects of training experiences and long-term interventions on mental state and EEG activity in athletes need further investigation in future studies.

1. Introduction

A growing body of recent research shows increasing attention on mental health in athlete populations (Reardon et al., 2021; Schinke et al., 2021). Data from previous studies have demonstrated that athlete populations have significant levels of mental health problems related to inevitable stressors such as overtraining, competition, injury, social expectation and media (Kuettel & Larsen, 2020; Rice et al., 2016), especially those elite athletes who are injured, in their later careers, retired or experiencing poor performance (Rice et al., 2016). Additionally, Pluhar et al. (2019) indicated that negative mental states are more common in individual sports athletes (e.g., track and field) than those who are in team sports (e.g., soccer). Given that, early prevention

interventions have been suggested to reduce the risk of mental health problems (Schinke et al., 2021), and to prevent mental health problems from affecting the training and competition of athletes. Adopting appropriate coping strategies is necessary to reduce the adverse effects of the stress (Hardy et al., 1996; Woodman & Hardy, 2003).

Mindfulness-based intervention (MBI) and psychological skills training (PST) are common strategies to cope with unwanted mental states (e.g., anxiety, worry) and increase focus on current tasks in sports (Gross et al., 2018; Josefsson et al., 2019; Röthlin et al., 2020). Mindfulness is generally described as a trait or state that referring to how an individual focuses intentionally and non-judgmentally on moment-by-moment experiences (Lindsay & Creswell, 2017). MBI refers to an educational and practical program that aims to foster mindfulness

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through an assortment of exercises including focused breathing, meditation and body scan (Creswell, 2017). MBI has been found to be effective for a large array of mental health issues, such as anxiety, stress and depression in non-athlete populations (Strohmaier, 2020), and improving negative rumination (Wu et al., 2021), eating concerns and psychological distress (Gross et al., 2018) and psychological well-being (Myall et al., 2022) in athlete populations. Moreover, there is also neurocognitive evidence that MBI alters brain activities involved in mental health (Lomas et al., 2015), as well as improving cognitive processing such as attention and executive control (Sumantry & Stewart, 2021).

PST, which is an umbrella term that consists of self-talk, imagery, goal setting and relaxation techniques (or arousal-regulation), is another approach often compared with MBI. In contrast to MBI, which involves focusing on experiences in the moment in an open and accepting way when athletes face positive or negative internal and external stimuli, (Röthlin et al., 2020), PST techniques usually emphasize self-control to create an ideal mental state (e.g., concentration, relaxation) by suppressing any unwanted thoughts (e.g., anxiety) or emotional states (e.g., negative affect) during performance (Gardner & Moore, 2004; Moore, 2009). PST has been widely adopted by sport psychologists to develop athletes' abilities to cope with stress and reduce unwanted thoughts (e.g., anxiety) or undesired emotional states (e.g., negative affect) (Hardy et al., 1996; Röthlin et al., 2020). While there are studies that have been demonstrated MBI to be more effective for health-related outcomes such as anxiety and emotional regulation, compared with PST (Gross et al., 2018; Josefsson et al., 2019; Röthlin et al., 2020), but most studies often used combined PST (i.e., at least two or more techniques) to compare the effect of PST in athletes to MBI (Gross et al., 2018; Josefsson et al., 2019; Röthlin et al., 2020), which might ignore or reduce the unique effects of an isolated PST (e.g., relaxation technique).

Relaxation techniques are classical PST techniques used frequently to cope with daily and competitive anxiety in elite athletes (Kudlackova et al., 2013). Relaxation techniques aim to induce a relaxation response and change internal thoughts by system breathing or guided imagery (Luberto et al., 2020). Several studies demonstrated that relaxation technique have comparable effects on mental health with MBI (Luberto et al., 2020; Manzoni et al., 2008; Rooks et al., 2017; Steyn et al., 2016). For example, Rooks et al. (2017) found that more engagement in relaxation training and mindfulness training are both associated with reduced anxiety and increased positive affect in athletes during high-demand intervals. Another study reported similar results that both relaxation and mindfulness training had benefits on psychological well-being and anxiety in music students (Steyn et al., 2016). Although the two practices have similar positive effects on mental health outcomes, Luberto et al. (2020) suggested that the similar effects are respectively through paying attention to present moment internal and external thoughts and without judgment (i.e., MBI), or eliciting relaxation responses to directly change internal events (i.e., relaxation techniques). This difference in mental processing might also be reflected in neural activity, as a longitudinal study in neuroimaging conducted by Sevinc et al. (2018) found that both MBI and relaxation-based intervention are associated with reduced stress, but MBI showed enhanced functional connectivity in brain regions involving sensory awareness and salience, whereas relaxation-based intervention showed enhanced functional connectivity in brain regions involving intentional inhibition and control. Given the similar benefits in mental health and different mechanisms of MBI and relaxation techniques, further comparing the underlying neural mechanisms in athletes during the implementation of the two practices is needed to provide greater insight into how sports practitioners and athletes appropriately use these psychological interventions for self-regulation in sports environments.

Moreover, based on the results of enhanced attention in experienced meditators, Gardner and Moore (2004) argued that the different implementation styles of MBI and PST might lead to different neural processing (e.g., allocation of cognitive resources) and distinct mental states. Specifically, MBI might allow for more efficient allocation of cognitive resources via automated attentional self-regulation and non-judgmental awareness in present moment, whereas PST instructs practitioners to focus on attentional control to suppress or alter external demands and internal experiences (Gardner & Moore, 2004; Moore, 2009). Although, efficient neural processing has been found to be influenced by continued practice of MBI (Tang et al., 2019), whether mental processing differs between MBI and isolated PST in novice mindfulness meditators is still unknown. Indeed, understanding the temporary response of individuals during these practices might have practical value and facilitate development of predicted measures, especially novice samples (Lin et al., 2020). Therefore, examining neural processing during a single bout of MBI and PST might help understand how athletes use these strategies to optimize their mental state, and provide valuable information for practitioners in sport psychology.

Mindfulness induction (MI), which is defined as a single bout of MBI within a brief period (e.g., five to 30 min), aims to induce a temporary state of mindfulness (Creswell, 2017; Howarth et al., 2019). MI might be a promising approach to examine neural processing during mindfulness exercise because it allows researchers to implement standardized methodological procedures in laboratory settings (Gill et al., 2020; Lin et al., 2020). During MI, participants are usually asked to deliberately bring their attention to an object or experience in the present moment without judgment (Gill et al., 2020). Accumulating evidence suggests that MI could improve mental health outcomes, including reducing anxiety (Westenberg et al., 2018), maintaining positive affect (PA; Hsu & Forestell, 2021), reducing negative affect (NA; Schumer et al., 2018). Several benefits of MI such as improving flow experience (Perry et al., 2017) cognitive functions (Zhu et al., 2022), reduced state anxiety (Perry et al., 2017; Wolch et al., 2020) and mental fatigue (Zhu et al., 2020) were also found in athlete populations. As such, MI might be an effective self-management approach to improve mental health in athletes even if its effects are considered relatively small and transient (Creswell, 2017; Howarth et al., 2019).

A single bout of relaxation is considered a relaxation induction (RI) in present study. Both mental relaxation techniques (e.g., autogenic relaxation, guided imagery) and somatic relaxation techniques (e.g., progressive muscle relaxation) aim to induce relaxation responses to reduce physiological and psychological stress (Luberto et al., 2020). RI has been found to produce a positive effect on mental health-related outcomes. For example, Hashim and Hanafi Ahmad Yusof (2011) found that a 30-min relaxation induction of either autogenic relaxation or progressive muscle relaxation both reduced negative mood states such as confusion, depression and tension in adolescent soccer players. Additionally, a study comparing the influence of MI and RI on state anxiety and mood suggested that both MI and RI reduce cognitive and somatic symptoms of anxiety, and negative mood (Lancaster et al., 2016). These empirical findings demonstrate that RI and MI have comparable benefits on mental health.

Electroencephalography (EEG) has been widely used to explore dynamic change and responses of the brain to detect the mental state changes in real-time (Kim et al., 2013). Neural oscillation of brain activity within specific ranges of frequency bands (e.g., theta, alpha) could reliably represent different mental processes (Lin et al., 2020). A systematic review that examined mindfulness meditation and EEG oscillations (Lomas et al., 2015) revealed that enhanced theta (4–8 Hz) and alpha (8–12 Hz) power are more often associated with mindfulness meditation as compared with eyes-closed resting state, mostly occurring

separately in the frontal region (e.g., prefrontal cortex) and posterior region (e.g., parietal cortex) (Lagopoulos et al., 2009). Theta power is usually considered a hallmark of situations that elicit anxiety and demand cognitive control (Cavanagh & Shackman, 2015), whereas alpha represents a low arousal, restful and relaxed state (Tang et al., 2019). Lomas et al. (2015) suggested that enhanced theta power and alpha power involve a unique brain activation pattern reflecting a state of relaxed alertness for mindfulness meditation. In addition, the increased theta power (Jacobs & Friedman, 2004) and alpha power (Mikicin & Kowalczyk, 2015) were also found following relaxation intervention, and during RI (Knott et al., 1997). Suggesting neural activity elicited by MBI and relaxation technique is comparable and similar. Nevertheless, there is still no research comparing directly different brain activity patterns during the MI or RI.

The purpose of this study was to examine the effect of brief MI and RI on anxiety and affect in athletes from individual sports, and to identify differences in brain activation between MI and RI using EEG (i.e., theta, alpha). The present study used a within-subject comparison across MI, RI, and control conditions (i.e., resting state). Considering MBI and PST might have similar effects on mental states in athletes (Röthlin et al., 2020) and non-athletes (Sevinc et al., 2018), it was hypothesized that MI and RI would both improve anxiety and affect, compared with a control condition. However, given that MBI and PST may exhibit different allocation of cognitive resources (Gardner & Moore, 2004; Moore, 2009), it was also hypothesized that RI would elicit greater theta power (i.e., more cognitive control) than MI. In addition, when compared with control condition, both MI and RI would elicit greater alpha power.

2. Methods

2.1. Participants

Thirty-five track and field athletes ($M_{\text{age}} = 20.63 \pm 2.43$ years, 27 male, 8 female) were recruited from two universities in Taiwan (see Table 1). The minimum sample size was estimated from an a priori power analysis using G*Power (power = 0.90, alpha = 0.05, effect size $f = 0.37$); a sample size of least 24 would adequately power a 3 (condition) by 2 (time) repeated-measures ANOVA interaction term (Faul et al., 2009). This effect size was based on previous work for the effect of brief MI on state anxiety in athletes (Wolch et al., 2020).

All participants were university-level track and field athletes who were competing in national and regional competitions, had engaged in sports training for an average of 8.57 ± 2.45 years, and had an average education of 14.86 ± 1.33 years. The specific athletic events are respective from sprints ($n = 4$), middle- and long-distance events ($n = 27$), hurdling ($n = 2$) and long jump ($n = 2$). The participants met the following inclusion criteria: (a) right-handed; (b) no history of neurological illness; (c) no regular training experience in mindfulness or relaxation interventions. Participants were provided written informed consent before the formal experiment. The study design and outcomes were registered on Clinical Trials (ID: NCT05452629). The protocol and consent forms were also approved by the authorized Research Ethics Committee of National Taiwan Normal University.

Table 1
Participant demographic information ($M \pm SD$).

Variable	Male	Female	Overall
Sample size (n)	27	8	35
Age (years)	20.56 ± 2.50	20.88 ± 2.30	20.63 ± 2.43
Education level (years)	14.70 ± 1.27	15.38 ± 1.51	14.86 ± 1.33
Sport experiences (years)	8.37 ± 2.60	9.25 ± 1.83	8.57 ± 2.45

2.2. Measures

2.2.1. State anxiety

The Chinese version of the State-Trait Anxiety Inventory (Wang & Chung, 2016), which is based on the State-Trait Anxiety Inventory (STAI, Spielberger, 1970), was used to assess state anxiety before and after each experimental manipulation in the current study. The State Anxiety Inventory (SAI) is a 20-item sub-scale in the STAI, with a 4-point Likert scale ranging from 1 (not at all) to 4 (very much so) for each item. Higher total scores indicate higher levels of state anxiety. Cronbach's alpha and McDonald's omega were $0.69 \sim 0.81$ and $0.53 \sim 0.75$ for SAI at pre-test and post-test across the three conditions.

2.2.2. Affect

The Positive and Negative Affect Schedule (PANAS) was used to assess positive affect (PA) and negative affect (NA) before and after each experimental condition. PANAS is a 20-item questionnaire that consists of two 10-item scales assessing positive affect state (i.e., PA) and negative affect state (i.e., NA), respectively. Participants rate to what extent they feel a certain way right now from 1 (not at all) to 5 (extremely). Higher total scores indicate higher levels of affect state. The internal consistency reliability of the positive subscale and negative subscale were satisfactory at pre-test and post-test across the three conditions (Cronbach's α for positive affect = $.87 \sim 0.91$) and (Cronbach's α for negative affect = $0.85 \sim 0.91$), respectively. Cronbach's alpha and McDonald's omega were $0.87 \sim 0.91$ and $0.88 \sim 0.91$ for the positive subscale, respectively, as well as $0.85 \sim 0.91$ and $0.86 \sim 0.91$ for the negative subscale at pre-test and post-test across the three conditions.

2.3. Procedures

The current study employed a within-subjects repeated measures design. All participants visited the laboratory on three occasions to complete three counterbalanced order conditions, including two experimental conditions (i.e., mindfulness induction, relaxation induction) and a control condition. The interval of each condition was at least one week. To ensure the MI and RI would be effective, prior to the first treatment session, all participants attended a 60-min orientation presented by a certified sport psychologist. The lecture included the theories of mindfulness and relaxation, application in sport environments, and two brief 5-min exercises involving formal mindfulness practice (e.g., becoming aware of the movement and sensations of the body when you are breathing naturally without judgment) and relaxation practice (e.g., imaging the muscles going from tension to relaxation using diaphragmatic breathing).

During the first visit, participants completed the informed consent form and demographics questionnaire, and were randomly assigned to attend 30 min of mindfulness (i.e., MI), relaxation (i.e., RI) or control conditions. To minimize order effects, these conditions were conducted in a counterbalanced order. Before each experimental condition, participants first completed the SAI and the PANAS. Then they were invited to a soundproof room, and guided in one of the three experimental conditions, while the electroencephalography (EEG) was recorded from beginning to end. Participants were instructed to close their eyes during the whole intervention to control for differences in EEG activity between eyes-closed and eyes-open conditions. After each experimental condition, participants immediately completed the SAI and the PANAS again. In total, the whole procedure lasted 50–60 min. Participants were monitored during each experimental condition and reminded to be awake when they appeared to be sleeping or distracted. All participants received compensation of approximately US \$100 for full participation in the three days of study.

2.4. Intervention conditions

During each experimental condition, participants were in a sound-proof room, asked to close eyes and keep a straight upright sitting posture, with muscles of face, neck, and shoulders relaxed, and hands resting on the lap and feet on the floor. For both the MI and RI conditions, participants listened to a 30-min audio recording of a female voice in Chinese.

2.4.1. Mindfulness induction

In the MI condition, participants were guided through the three classic mindfulness exercises sequentially (i.e., focused breathing, mindful meditation and body scan). Specifically, focused breathing was delivered at the first 10 min of the audio, and participants in this segment were guided to pay attention to the feeling of air coming into or out of their nose while breathing, whatever they are thinking at the moment; The next 10-min segment was mindful meditation, and participants were guided to monitor their thoughts and physical sensations in an open, nonjudgmental way. Lastly, body scan was delivered in the last 10 min of the audio. This segment instructed participants to direct their awareness toward each body area and observe or feel them at the moment. Participants were instructed to focus on the present moment by a depth breathing with a non-judgmental manner, and to just gently refocus on the sensations of breathing or body when their mind wander to something other than the present moment (e.g., daydream).

2.4.2. Relaxation induction

In the RI condition, participants were guided to relax each muscle group following the audio for three cycles. To match the absence of physical movement in MI protocol, the combination of self-directed relaxation exercise (Bernstein et al., 2000) and passive progressive muscle relaxation (Feldman et al., 2010) were used to induce relaxation. The exercise first asked participants to use diaphragmatic breathing six to eight times. Participants were then guided to sequentially notice sensations of tension in their muscles from head to feet, and keep breathing with slow and paced breaths. Specifically, participants were instructed to relax these muscles, and imagine the tension going away with every breath, and finally observe these muscles becoming deeply relaxed. In the end, participants were asked to count their breathing slowly 15 to 20 times and then count down from 4 to 1 to end a cycle. Participants repeated this process three times (10 min for a cycle).

2.4.3. Control condition

In contrast to the MI and RI conditions, participants received no information about breathing or thought guidance. Instead, they were just informed to sit relaxed and allow their mind to wander freely (e.g., recall past or future experiences in daily life).

2.5. EEG recoding and analysis

2.5.1. EEG recoding

Continuous EEG activity was recorded during three 30 min experimental conditions in a quiet room using a Neuroscan elastic cap with 32 passive Ag/AgCl electrodes (Neuroscan Quick-Cap, Neuroscan Inc., VA) based on the International 10/20 system, the montage was referenced to the average of the right and left mastoids and ground electrode placed on Fpz site (Jasper, 1958). To remove ocular artifacts and eye movement from the EEG data, horizontal and vertical electrooculograms (VEOG and HEOG) were also recorded by electrodes placed above and below the left eye as well as next to both eyes. All electrode impedances were maintained below 10 k Ω before the start of the recording. The EEG and EOG data were sampled with 500 Hz and amplified using Neuroscan Synamps2 amplifier (Scan 4.5, Neurosoft Labs, Inc.), with a band pass filter set at 0.1–100 Hz, and the notch filter at 60 Hz during the data collection.

2.5.2. EEG analysis

Offline analyses were performed using Neuroscan Edition 4.3 software. The EEG signals were re-referenced to the average of the left and right mastoids. Ocular artifacts were corrected using a band-pass cutoff of 1–30 Hz (12 dB/octave) and EOG corrected by an algorithm (Semlitsch et al., 1986) to eliminate the effects of blinks and eye movement. To analyze the cortical activity in the three experimental conditions, the analysis interval was segmented into epochs of 1s, which would reject amplitudes that exceeded ± 100 μ V from subsequent analyses (mean percentage of epochs rejected = 6.29%; SD = 5.78) (Wang et al., 2020).

Mean power spectral densities (μ V²/Hz) were computed across all epochs within each experimental condition through a fast Fourier transform (FFT), these artifact-free segments (length 1s) were transformed from the temporal to frequency domain after data were tapered at the distal 10% of each epoch by Hanning window. Next, the natural logarithm transformation was conducted in Excel to compute a normal distribution of power values (Lin et al., 2020). The selected frequency bands were theta (4–8 Hz) and alpha (8–12 Hz). The frequency bands were computed by averaging across epochs and integrated spectral power in the frontal region (Fp1, Fp2, F7, F3, Fz, F4, F8) for theta power, and in the posterior region (P7, P3, Pz, P4, P8, O1, Oz, O2) for alpha power. The selected brain regions were based on previous studies that suggested theta power and alpha power usually occur maximally in the frontal region and posterior region, respectively (Lagopoulos et al., 2009; Lin et al., 2020).

2.6. Data analysis

Data analyses were performed using the SPSS for Mac 26.0 software with alpha value set at 0.05. The protocol employed a repeated measure design with the experimental conditions and time points as within-subject factors. A 3 (conditions: MI vs. RI vs. control) $\times$ 2 (time: pre-test vs. post-test) repeated measures analysis of variance (ANOVA) was conducted for the SAI and the PANAS, respectively, to test the effects of brief MI and RI on state anxiety and affect. For the EEG data, separate one-way ANOVAs were conducted to compare theta power and alpha power during the three conditions. The Greenhouse-Geisser correction was applied where sphericity was violated to control possible Type I error and multiple comparisons were conducted using Bonferroni-adjusted if there were any significant effects. Estimated effect sizes were reported using partial eta squared (η_p^2). Mean and standard deviation were presented.

3. Results

3.1. Mental outcomes

3.1.1. State anxiety

Two-way ANOVA revealed a time effect for SAI score [$F(1, 34) = 17.11, p < .001, \eta_p^2 = 0.34$], which was superseded by a significant condition $\times$ time interaction [$F(2, 68) = 6.48, p = .004, \eta_p^2 = 0.16$]. Post hoc analysis indicated that the decreased scores from pre-test to post-test in MI (37.83 ± 10.82 vs. $31.31 \pm 9.44, p < .001$) and RI (36.94 ± 9.51 vs. $32.00 \pm 8.66, p < .001$), but not control condition (34.86 ± 8.03 vs. $34.06 \pm 9.40, p = .536$). However, no significant condition effect was observed for SAI scores ($p = .996$) (see Figure 1a).

3.1.2. Affect

Two-way ANOVA revealed no significant time effect, condition effect or condition $\times$ time interaction effect on the positive subscale of PANAS ($ps > .05$, see Figure 1b). For the negative subscale of PANAS, a significant time effect was observed [$F(1, 34) = 28.79, p < .001, \eta_p^2 = 0.46$], which was superseded by a significant condition $\times$ time interaction [$F(2, 68) = 4.03, p = .022, \eta_p^2 = 0.11$]. Post hoc analysis indicated significant decreases from pre-test to post-test in MI (17.80 ± 7.69 vs. $13.63 \pm 6.83, p < .001$) and RI (16.80 ± 6.82 vs. $14.40 \pm 6.59, p < .001$), but no

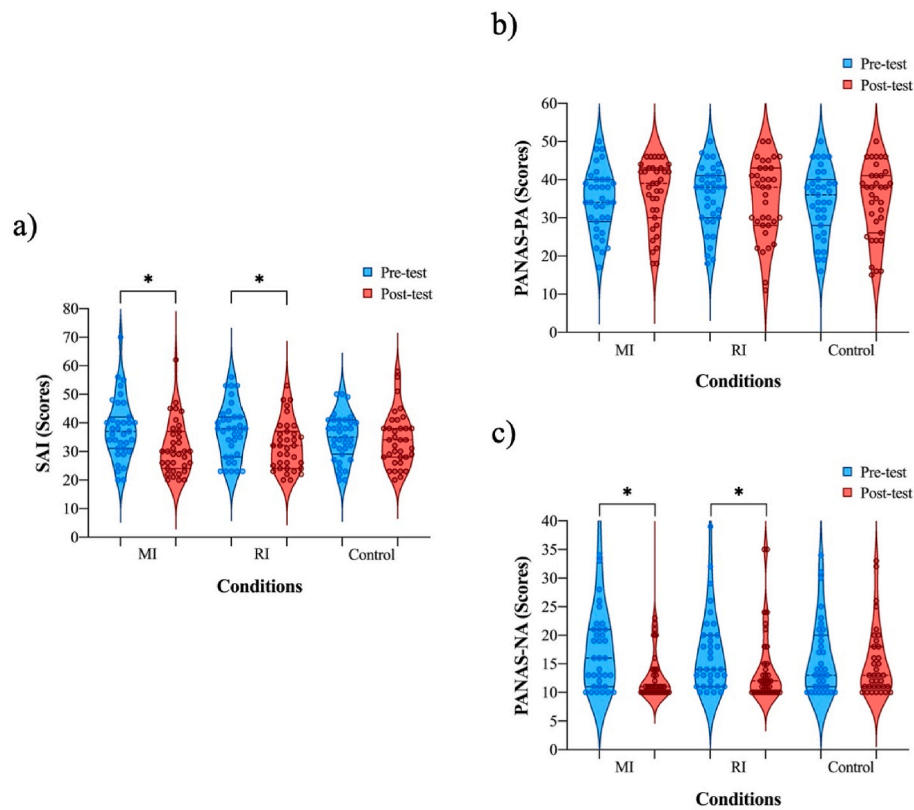


Figure 1. Violin plots demonstrating the changes in (a) SAI, (b) PANAS-PA, and (c) PANAS-NA scores for each of the experimental conditions. *Significant difference within the condition; SAI = State Anxiety Inventory; PANAS = Positive and Negative Affect Schedule; PA = positive affect; NA = negative affect; MI = mindfulness induction; RI = relaxation induction.

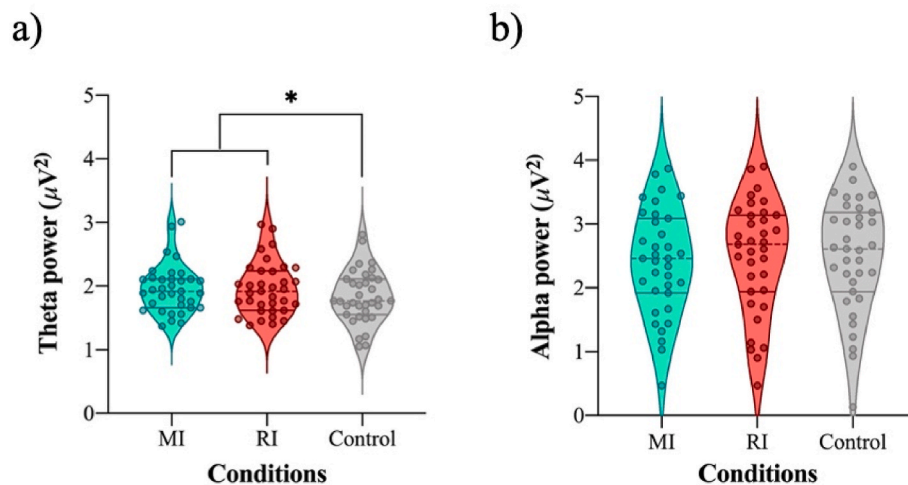


Figure 2. Violin plots demonstrate the activity of (a) frontal theta power (4–8 Hz), and (b) posterior alpha power (8–12 Hz) during each experimental condition. *Significant difference between the condition; MI = mindfulness induction; RI = relaxation induction.

change in the control condition (15.86 ± 6.55 vs 15.20 ± 6.11 , $p = .400$). However, no significant condition effect was observed for NA scores ($p = .981$) (see Figure 1c).

3.2. Electrophysiological outcomes

One-way ANOVA revealed a significant difference for theta power among the three experimental conditions [$F(2, 68) = 5.10$, $p = .009$, $\eta_p^2 = 0.13$]. Post hoc analysis indicated that compared with control condition (1.83 ± 0.42 Ln μV), participants had a higher frontal theta power

during MI (1.96 ± 0.38 Ln μV , $p = .026$) and RI (1.96 ± 0.41 Ln μV , $p = .028$), but no significant difference in frontal theta power between MI and RI ($p > .05$) (see Figure 2a). Additionally, One-way ANOVA revealed no significant difference for posterior alpha power among the three experimental conditions [$F(2, 68) = 1.24$, $p = .296$] (see Figure 2b). The topographic distributions of the theta power and alpha power across three experimental conditions are also presented in Figure 3.

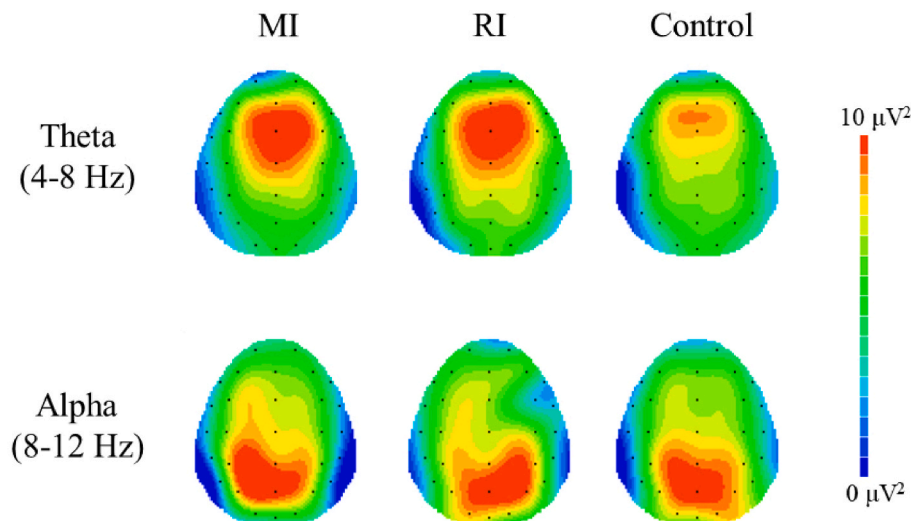


Figure 3. Topographic scalp distribution of the frontal theta (4–8 Hz) and posterior alpha (8–12) band power during the different conditions. MI = mindfulness induction; RI = relaxation induction.

4. Discussion

The present study examined the effect of MI and RI on anxiety and affect in individual sport athletes. In addition, we sought to identify differences in EEG activity between MI and RI. The primary results revealed that both MI and RI significantly reduced negative affect states (i.e., state anxiety & NA) across time compared with the control condition, but no differences in positive affect were observed. On the other hand, inconsistent with our hypothesis for EEG activity, participants exhibited higher frontal theta power during MI and RI compared to the control condition, but no differences in posterior alpha power during the three conditions. These findings suggest that decreased state anxiety and negative affect during MI and RI inductions might be associated with greater attention control.

The present finding is that MI and RI provide similar improvements in negative mental states. In terms of MI, the current results fall in line with the findings of mindfulness studies in upper extremity patients (Westenberg et al., 2018) and college students (Perry et al., 2017), showing single-session mindfulness intervention could have an immediate benefit on unwanted mental states such as anxiety, depression or anger (Perry et al., 2017; Westenberg et al., 2018). Similarly, current findings revealed decreased state anxiety following RI, which indicates that a 30-min self-directed relaxation is also an effective strategy for coping with daily anxiety (Kudlackova et al., 2013). Furthermore, the current study found that athletes had reduced negative affect following both MI and RI, which is consistent with Vinci et al. (2014) and Lancaster et al. (2016) who reported similar improvements in negative affect following brief MI and RI in college students. Similarly, Lee and Orsillo (2014) observed that participants with generalized anxiety disorder showed decreased state anxiety after MI, RI, and even a control condition, but improvement in cognitive flexibility was found only following MI and RI. They suggested that this greater improvement in anxiety and cognitive flexibility following MI and RI might be caused by increased vagal tone. It could also be linked to a result from an induced relaxation response (Luberto et al., 2020). A common physiological state of relaxation could be caused by mindfulness meditation and relaxation techniques (Vinci et al., 2014). However, relative to relaxation techniques that aim to elicit relaxation response to counter the stress response, mindfulness practices tend to improve mental health through inducing awareness and acceptance toward moment-by-moment experiences, and do not depend completely on relaxation response (Luberto et al., 2020). For instance, Lindsay et al. (2018) conducted pre-registered dismantling trials to examine acceptance skills as a

potential mechanism underlying MBI's reduction of biological stress reactivity after acute stress manipulation. They found that MBI with an acceptance component was more effective in reducing biological stress reactivity than MBI without an acceptance component. The finding suggests acceptance might be a unique mechanism by which MBI improves mental health when individuals face stressors and negative experiences. Regardless, current findings suggest that brief MI and RI are ways to manage negative mental states in athletes, but more research is needed to explore the slight difference in potential mechanisms between MI and RI effect on mental health.

Although MI and RI may have similar beneficial effects on negative mental states (i.e., anxiety, NA), PA did not differ among the three conditions or from pre-test to post-test. A recent systematic review explored the effect of brief MBI in a laboratory context (Jiménez et al., 2020), suggesting that the effect of brief MBI on PA is contradictory. For example, Hsu and Forestell (2021) found a protective effect of a 15-min MI on reduced PA induced by negative mood manipulation, compared with a control group. In contrast, other studies found that PA was reduced (Lancaster et al., 2016; Vinci et al., 2014) or there was no significant change (Somaraju et al., 2021) following MI. Some possible explanations for the contradictory findings are that first, the 30-min MI and RI might not be strong enough to alter PA; that is, a brief intervention would be unlikely to have as strong a beneficial effect as long-term interventions. Therefore, more MBI program engagement is needed to increase the PA (Rooks et al., 2017; Vinci et al., 2014). Second, the athletes tested in the present study had a slightly higher PA score at baseline (i.e., baseline PA score across conditions = 34.31–34.57) than the non-athlete populations in other studies (Lancaster et al., 2016; Vinci et al., 2014), suggesting a possible ceiling effect. The results from Roeh et al. (2020) might support this possibility. They demonstrated that marathon runners report fewer depressive symptoms and more positive affect than sedentary people, and further observed that negative affect continued to decrease 24 h after long-distance running in marathon athletes, whereas positive affect remained unchanged. If athlete populations already have relatively higher PA, that might partially explain why PA was unaffected by MI or RI in this study.

Due to the difference in rationale and implementation between MBI and PST (Gardner & Moore, 2004), we expected different neural processing in MI, RI, and control conditions. However, current findings are contrary to this hypothesis. Participants showed significantly greater theta power during MI and RI than during the control condition, but no significant difference between MI and RI. Specifically, increased frontal theta power involves internally-directed attention that is used to

develop top-down control during meditation (Brandmeyer & Delorme, 2018), and it has been found to accompany enhanced cognitive control and awareness (Cavanagh & Shackman, 2015; Tang et al., 2019). Furthermore, this cortical activation in anterior brain regions during RI is considered a central nervous system-relevant relaxation response induced by repetitive mental focus after individuals receive relaxation instructions (Jacobs et al., 1996; Jacobs & Friedman, 2004; Knott et al., 1997). Considering that improved negative mental states are observed only after MI and RI, increased frontal theta might also link to the improved negative mental state of athlete participants in this study, as McEvoy et al. (2017) suggested that reductions in cognitive anxiety appear to reflect the process of rising awareness and redeployment of attention during interventions that combine mindfulness practice and progressive muscle relaxation. It is possible that increased frontal theta power during the MI and RI is the result of cognitive processing involving effortful self-regulation in athletes who are novice meditators.

Though results revealed greater frontal theta power in MI and RI than in the control condition, there was no difference in allocating cognitive resources between MI and RI, which conflicts with past research on differences in allocation of cognitive resources between MBI and PST (Gardner & Moore, 2004; Moore, 2009). This might be attributed to modulating effect of mindfulness experiences. Brandmeyer and Delorme (2018) suggested that experienced meditators show a greater depth and frequency of sustained meditation than novice meditators. Further, a previous study reported that compared with novice meditators, experienced meditators had stronger deactivation in prefrontal cortex during mindfulness meditation (Lutz et al., 2016). These results indicate that Gardner & Moore's (2004) assumption might only be applicable to experienced meditators but not novices. Current findings revealed similar cognitive processing represented by increased frontal theta power in novice meditators during the MI and RI. Given that state changes in oscillatory activity would be influenced by continued mental practice (Brandmeyer & Delorme, 2018; Cahn et al., 2010) and differences in long-term neural effects between MBI and relaxation-based intervention have been found in a recent neuroimaging study (Sevinc et al., 2018), further exploring the neural effects of long-term mindfulness and relaxation interventions in athletes might clarify the potential benefits of the two psychological interventions.

Contrary to expectations, results did not reveal differences in posterior alpha power among the three conditions. This finding is similar to previous observations with both experienced meditators (Berkovich-O'hana et al., 2014; Cahn et al., 2010) and novices (Lin et al., 2020); namely, no differences in alpha power between mindfulness and resting state were found. This unexpected result could be attributed to participants feeling somewhat relaxed during the control condition, as they were instructed to think about anything and be relaxed. Indeed, past work in neuroscience has been found increased posterior alpha power during the resting state (Scheeringa et al., 2012). Furthermore, the resting state might be associated with more mind-wandering, which was correlated with higher alpha power during a lengthy cognitive task, compared to participants who reported concentrating on the task (Compton et al., 2019). These findings provide a possible explanation for similar alpha power among the MI, RI and control conditions. To better understand and get a clear picture of changes in theta and alpha power during the brief psychological intervention, additional studies will be needed to explore whether prior mindfulness experience modulates these effects (Cahn et al., 2010; Lin et al., 2020), and to consider the potential effect of mind-wandering on EEG activities during psychological intervention (Compton et al., 2019).

To our knowledge, this is the first study to directly compare the effect of MI and RI on anxiety, affect, and brain activation using EEG in athletes. The present study, nevertheless, has some limitations that need to be considered. First, participants in the control condition showed higher pre-test scores of state anxiety and negative mood, while there was no statistically significant difference between conditions. Despite adopting a counterbalanced design, anxiety and affect are susceptible to

interference by daily events that might limit the effectiveness of inductions. To minimize the possibility that significant interactions might be driven by higher pre-test scores, future investigations might use questionnaires to screen specific situations (e.g., competitive period, examination) that affect participants' mood state to ensure mental states are equated between conditions at baseline. The second limitation is that relaxation was the only PST compared with mindfulness in the current study. Other PST techniques, such as imagery or self-talk, may lead to different mental states and EEG activity. Consequently, examining the differential effect between mindfulness and other PST in future studies is needed. Third, this study used EEG frequency domain analysis to observe the alternation of theta power and alpha power during the mindfulness and relaxation practice. Given the findings of past research on meditation and mindfulness training (Lomas et al., 2015), future research could continue to compare the difference between MI and RI in other EEG frequency bands (e.g., delta power, gamma power) and time-frequency domain in cognitive tasks (i.e., event-related potential) after brief intervention. Fourth, with the absence of manipulation checks in the study we cannot be that certain participants were engaged in MI or RI. According to previous neurocognitive research in mindfulness and relaxation, respectively (Brandmeyer & Delorme, 2018; Knott et al., 1997; Sevinc et al., 2018), we assumed that EEG activity in each condition would show different patterns and did not carry out manipulation checks. Unfortunately, current findings did not show a predicted difference in neural activity patterns between MI and RI, which might limit the interpretation of current findings. Lastly, results for anxiety should be interpreted cautiously given that the SAI had relatively lower internal consistency at pre-test and post-test across the three conditions (McDonald's omegas = .53 ~ .75). While Cronbach's alpha showed relative acceptable range (i.e., .69 ~ .81), McDonald's omegas have been suggested to provide more reliable information about adopted measures (Watkins, 2017). Future research examining the effect of MI and RI on anxiety measured by SAI should ensure adequate internal consistency to confirm the results in the present study.

5. Conclusion

The current study provides initial evidence that brief MI and RI both improve negative mental states (i.e., decreased anxiety and NA) in individual sports athletes who do not have regular training experience in mindfulness or relaxation interventions, and both exhibit similar neural mechanisms (i.e., higher frontal theta power) during MI or RI. These results imply that a single bout of MI or RI have comparable psychological benefits through similar neural processing for athletes, compared with the control condition. Suggesting athletes could choose the most appropriate interventions based on their preference to achieve an ideal mental state in daily life or competition, especially those athletes who are mindfulness/relaxation novices.

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Declaration of competing interest

Given their roles as an Editorial Board Members, Hung T.-M. and Chang Y.-K. had no involvement in the peer-review of this article and had no access to information regarding its peer-review. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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